

Method for Predicting Odor Intensity of Perfumery Raw Materials Using Dose–Response Curve Database

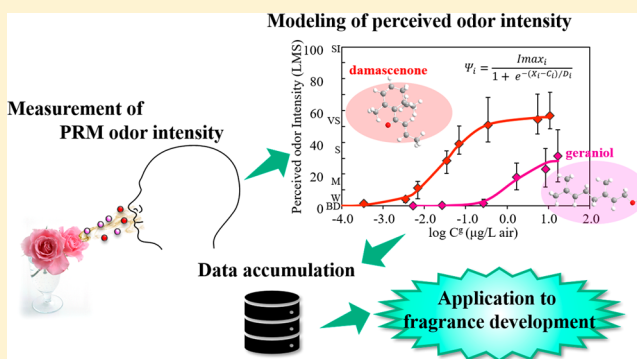
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Supporting Information

ABSTRACT: The main purpose of this study is to facilitate fragrance development on the basis of scientific knowledge. To this end, data on 314 perfumery raw materials (PRMs) showing the relationship between PRM odor intensity and gas concentration were obtained, and a calculation model for the data was then developed with the following features: (1) maximum PRM coverage, (2) calculating values implying odor intensity from only arbitrary gas concentration, and (3) estimating odor intensity from the calculated values directly and easily. To verify the prediction accuracy of this model, the predicted odor intensity was compared with the evaluated value for both single component and a mixture, and the same degree of root-mean-square error (RMSE) was confirmed. RMSE in the single component was 6.22 while that in the mixture was 6.69. Thus, the odor intensity of a PRM or mixture can be predicted from arbitrary gas concentrations.



INTRODUCTION

Addition of fragrance is one of the ways to enhance the attractiveness of products (such as detergents, softeners, shampoos, and body washes) or communicate bland concepts to consumers in modern markets. In general, fragrances for products are formulated by perfumers who acquire extensive knowledge about perfumery raw materials (PRMs) and sophisticated skills through long-term experience. One of the problems in fragrance development is that it is difficult to predict the fragrance performance, such as odor intensity and character, from the prescription. The main cause of this difficulty is the difference in the PRM sensory characteristics. It is well-known, for example, that each odor molecule (including PRM) has a different threshold, i.e., the lowest concentration at which we can perceive the odor of the molecule, and slope, i.e., the ratio of odor intensity change to gas concentration change.^{1,2} Thus, it is impossible to estimate the PRM odor intensity from only its concentration in the gas phase. In addition, it is widely known that the balance of the fragrance character or odor intensity drastically changes under the influence of synergistic or offset action even though no more than 0.01% PRM is added to the prescription.³ As the details of these physicochemical and biological phenomena related to odor perception are unclear, a perfumer's craftsmanship and experience remain essential for fragrance development in spite of extensive progress in science and technology.

Realizing a technology for the quantitative prediction and explanation of odor character and intensity from each PRM physical parameter or amount remains a major challenge in this

field, and thus far, studies toward this end have been conducted with various approaches. Several groups, such as those of Rodrigues,^{4–9} Chastrette,¹⁰ Keller,^{11,12} Ihara,¹³ Kim,¹⁴ Saison,¹⁵ and Wu,¹⁶ have been actively involved in this study. In addition to the physical data of PRM and biological data regarding odor, data on the accumulation of numerous odor sensation concerning cognition, perception, and preference have been extensively collected and studied to enhance the precision of the prediction model.^{11,12,17}

Considering the recent studies on the prediction of odor intensity based on the viewpoint of chemical engineering, the report by Rodrigues's group is a useful example.^{4–9} They first tried to estimate each PRM gas concentration from some physical properties and then tried to predict the odor intensity through a developed calculation model that converts the gas concentration into odor intensity based on the theory of odor value (OV)^{18,19} and Stevens's law.²⁰ Finally, they confirmed a statistically good correlation between the results of the odor intensity predicted by the test model used by the P & G panelists and the evaluated values.⁷ However, we believe that this model has scope for improvement in at least two aspects. The first aspect is that the PRM threshold is necessary to estimate the odor intensity through their model based on OV theory. As the threshold varies considerably depending on the

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Table 1. Materials Information

CAS no.	compound	purity	supplier
5471-51-2	4-(4-hydroxyphenyl)-2-butanone (raspberry ketone)	>99%	Tokyo Chemical Industry Co., Ltd.
87-44-5	beta-caryophyllene	>90%	Tokyo Chemical Industry Co., Ltd.
41519-23-7	cis-3-hexenyl isobutyrate	>95%	Tokyo Chemical Industry Co., Ltd.
91-64-5	coumarin	>99%	Tokyo Chemical Industry Co., Ltd.
105-95-3	ethylene glycol brassylate (ethylene brassylate)	>95%	Tokyo Chemical Industry Co., Ltd.
142-92-7	hexyl acetate	>99%	Tokyo Chemical Industry Co., Ltd.
93-58-3	methyl benzoate	>99%	Tokyo Chemical Industry Co., Ltd.
119-36-8	methyl salicylate	>99%	Tokyo Chemical Industry Co., Ltd.
541-91-3	muscone	>97%	Tokyo Chemical Industry Co., Ltd.
111-87-5	<i>n</i> -octanol	>99%	Tokyo Chemical Industry Co., Ltd.
106-44-5	<i>p</i> -cresol	>99%	Tokyo Chemical Industry Co., Ltd.
43052-87-5	alpha-damascone	≥95%	Sigma-Aldrich
80-56-8	alpha-pinene	98%	Sigma-Aldrich
14901-07-6	beta-ionone	>96%	Sigma-Aldrich
18172-67-3	beta-pinene	≥99%	Sigma-Aldrich
928-96-1	cis-3-hexen-1-ol (<i>cis</i> -3-hexenol)	≥98%	Sigma-Aldrich
3681-71-8	cis-3-hexenyl acetate	≥98%	Sigma-Aldrich
5392-40-5	citral	≥96%	Sigma-Aldrich
23696-85-7	damascenone	≥98%	Sigma-Aldrich
106-72-9	2,6-dimethyl-5-heptenal (melonal)	80%	Sigma-Aldrich
140-88-5	ethyl acetate	≥99%	Sigma-Aldrich
105-54-4	ethyl butyrate	≥98%	Sigma-Aldrich
39255-32-8	ethyl 2-methylpentanoate (manzanate)	≥98%	Sigma-Aldrich
120-72-9	indole	≥99%	Sigma-Aldrich
125-12-2	isobornyl acetate	≥95%	Sigma-Aldrich
503-74-2	isovaleric acid	99%	Sigma-Aldrich
80-54-6	2-(4- <i>tert</i> -butylbenzyl)propionaldehyde (lilial)	≥96%	Sigma-Aldrich
78-70-6	linalool	>97%	Sigma-Aldrich
5989-27-5	limonene	≥99%	Sigma-Aldrich
115-95-7	linalyl acetate	≥97%	Sigma-Aldrich
127-51-5	3-methyl-4-(2,6,6-trimethyl-2-cyclohexen-1-yl)-3-buten-2-one (methyl ionone-gamma)	≥90%	Sigma-Aldrich
93-04-9	2-methoxynaphthalene (nerolin yara yara)	≥99%	Sigma-Aldrich
80-71-7	methyl cyclopentenolone	98%	Sigma-Aldrich
88-41-5	<i>o</i> - <i>tert</i> -butylcyclohexyl acetate (<i>o</i> -t-b.c.h.acetate)	≥99%	Sigma-Aldrich
60-12-8	phenylethyl alcohol	≥99%	Sigma-Aldrich
83-34-1	skatole	99%	Sigma-Aldrich
68039-49-6	2,4-dimethyl-3-cyclohexenecarboxaldehyde (triplal)	≥97%	Sigma-Aldrich
112-45-8	aldehyde c-111 len	>96%	Kao Chemicals Europe
110-41-8	aldehyde c-12 mna	>97%	Kao Chemicals Europe
1247790-47-1	3,6-dimethyl-2-heptanol (terpirosa)	>95%	Kao Chemicals Europe
706-14-9	gamma-decalactone	>98%	Kao Chemicals Europe
104-67-6	gamma-undecalactone	>98%	Kao Chemicals Europe
55066-48-3	phenyl hexanol	>97%	Kao Specialties Americas LLC
67801-64-3	floramat	>95%	Kao Specialties Americas LLC
25265-71-8	dipropylene glycol	99%	FUJIFILM Wako Pure Chemical Corporation
67-64-1	acetone	99.5%	FUJIFILM Wako Pure Chemical Corporation
8042-47-5	liquid paraffin	≥99%	FUJIFILM Wako Pure Chemical Corporation

measurement method, the value is different in each article.^{21,22}

In short, the value of the odor intensity calculated from the model is strongly influenced by the threshold value. The second aspect is the meaning of the value calculated from the model. The calculated value is only converted from the ratio of gas concentration to the threshold exponentially without a labeled semantic description. Therefore, it is considerably difficult to identify how strongly the value implies odor intensity or by how much the odor intensity increases with this value calculated from the model. In summary, for perfumers to use such a prediction technology more conveniently in their workplace, a model, which makes it possible to directly

calculate a more easily understandable value indicating odor intensity, is required.

To address the above-mentioned problems, the present study focuses on the development of a model with three notable features: (1) maximum coverage of the major PRMs for fragrance creation, (2) calculation of values implying odor intensity from only arbitrary gas concentration, and (3) estimation of odor intensity from the calculated values directly and easily. Specifically, the odor intensity of 314 types of PRMs generally used for creating fragrances with different gas concentrations at multiple points was evaluated with the labeled magnitude scale (LMS)^{23,24} by panels comprising

perfumers and evaluators who are actively engaged in fragrance creation; then, the experimentally obtained data were curve-fitted with a logistic function. Obtaining data that show the relationship between the PRM odor intensity and gas concentration and constructing an equation that yields the result make it possible to calculate the odor intensity of each of the 314 PRMs directly from arbitrary gas concentration data. Moreover, the calculated value permits interpretation of the odor intensity as semantic information through the LMS value used for quantification or comparison in studies concerning taste and smell. Thus, obtaining a large amount of PRM data, which are commonly used for fragrance creation, can facilitate the development of a wide variety of fragrances for products. This paper reports the data obtained in our study as well as the results of verification of the prediction accuracy of the developed calculation model.

■ EXPERIMENTAL SECTION

2.1. Materials. The materials used and their CAS no., purity, and supplier are listed in Table 1.

2.2. Measurement of PRM Gas Phase Concentration.
Sample Preparation. Liquid PRM or DPG (0.5 mL) mixed with a powdered PRM was spiked in a 3 L sample bag (GL Science Inc.). Then, the bag was sufficiently filled with air at room temperature through a filter, and the inlet was hermetically sealed with a silicone stopper. Next, the bag was maintained at room temperature for at least 12 h to attain an adequately dense PRM concentration in the gas phase. A syringe was used to collect air from a 3 L sample bag. The needle of the syringe was inserted into the silicone stopper and only sufficient air containing PRM gas was collected from the bag. Then, the collected air was injected into a 1 L fluororesin bag in a hermetical state. Subsequently, purified air at room temperature for diluting the gas sample was introduced into the fluororesin bag with another syringe. For preparing a 100-fold diluted sample, 10 mL of air containing the PRM was mixed with 990 mL of purified air in the 1 L fluororesin bag. To prevent the PRM in the gas phase injected in the bag from decreasing because of adsorption on the syringe surface, the sample gas was introduced into the syringe several times in advance. Finally, the gas concentration inside the prepared sample bag was quantified with gas chromatography–mass spectrometry (GC–MS).

GC–MS Analysis. All analyses were performed on a system comprising a gas chromatograph (6890N, Agilent Technologies) equipped with a thermal desorption system (TDS3, Gestel), thermal desorption autosampler (TDSA2, Gestel), modular analytical system (controller C506, Gestel), and a mass spectrometer (5975B, Agilent Technologies) as a detector. For collecting the aroma sample, Tenax TA glass was used as an adsorbent along with a sampling pump (SP208 Dual, GL Science Inc.). Tenax TA glass connected to the sampling pump was also connected to the prepared sample bag with a Teflon connector. The aroma in the bag was drawn at 50 mL/min with the sampling pump, and 100 mL of the aroma was finally adsorbed on the Tenax TA glass, which was set in TDSA2. For the TDS, the following parameters were used: 5 °C, held for 30 s; ramped to 250 °C (122 °C/min), held for 5 min. The TDS was operated in the splitless mode. Cooled Injection System (CIS) was programmed as follows: cooled to –150 °C with liquid nitrogen, held for 30 s, heated to 150 °C at 16 °C/min and then to 250 °C at 12 °C/min, and held for 10 min. The transfer line was maintained at 260 °C. Separation

was performed on a DB-WAX capillary column (length = 60 m, ID = 0.250 mm, d_f = 0.25 μ m; Agilent Technologies) with helium as a carrier gas at a constant pressure of 116.5 kPa and a flow rate of 1.1 mL/min. The oven was programmed as follows: 40 °C, held for 3 min, heated to 70 °C at 6 °C/min and then to 240 °C at 3 °C/min, and held for 20 min. Mass spectrometric scanning was performed in the range of 40–322 m/z at a rate of 2.58 scan/s. The repeatability of the GC/MS quantification was confirmed with 8 PRMs (floramat, raspberry ketone, ethylene brassylate, damascenone, linalool, terpirosa, citral, and gamma-decalactone) using above same method; the relative standard deviation (RSD) of the mean was approximately 3.6% (see Table S1).

2.3. Measurement of PRM Odor Intensity. Panelists. A group of 18 subjects [10 males and 8 females; age range, 23–39 years; median age, 32.3 (male) and 30.0 (female)] comprising evaluators and perfumers actively engaged in fragrance evaluation and creation participated in the measurement. In addition, all the subjects had the required aptitude for passing the panel selection test with five standard odorants,²⁵ namely, phenylethyl alcohol, methyl cyclopentenolone, isovaleric acid, gamma-undecalactone, and skatole. Two of the subjects had been awarded a national qualification as olfactory measurement operator by the Japan Association on Odor Environment designated by the Japanese Ministry of the Environment. Measurement of odor intensity was conducted by 3–5 panelists randomly selected from the above-mentioned group in accordance with the Japanese simple olfactory test method²⁶ to reduce the time and cost of measurement and to accumulate the data efficiently.

Olfactometer. Odor intensity measurement was performed using an olfactometer consisting of a fragrance and flavor autosampler (FAS-1, Shimadzu Corporation) and a fragrance and flavor diluter (FDL-1, Shimadzu Corporation).^{27–29} A 1 L fluororesin bag containing aroma quantified with GC/MS was set to FAS-1. Then, the sample gas was automatically introduced into FDL-1 from the bag with a 100 mL syringe included with FDL-1. Odorless air at room temperature filtered through a gas clean charcoal filter (CP17972, Agilent Technologies) was used as the carrier gas, and the flow rate was programmed to remain constant at 200 mL/min. The sample gas mixed with the aroma and carrier gas in the flow path was finally discharged from the Teflon tube (3.0 mm OD and 2.0 mm ID) at 350 mL/min. FDL-1 was programmed as follows: the sample gas was discharged for 4 s; odorless air was discharged for 4 s. After two sets of the cycle and discharge of the sample gas for 4 s, the device finally stopped. The test room was maintained in a quiet environment with a temperature range of 21–28 °C. Furthermore, the room was adequately ventilated and kept in an odorless condition at all times. All the prepared samples were evaluated on the day of preparation.

2.4. Evaluation Scale. The labeled magnitude scale (LMS) developed by B. G. Green, which is a hybrid scaling technique using a verbally labeled line with quasi-logarithmic spacing between the labels, was employed as the evaluation scale.^{23,24} The LMS is useful for quantifying and measuring the difference in perceptual intensity between subjects directly or in detail. Hence, it is widely used for quantification and comparison of sensational intensity concerning taste and smell. The LMS is a labeled verbal anchor with a range of 0–100, and the values denote the following: barely detectable, 1.4; weak, 6.1; moderate, 17.2; strong, 35.4; very strong, 53.3; and

strongest imaginable, 100. Thus, one can estimate the odor intensity to some extent from the LMS value. Each liquid paraffin, mixed with 0.80 wt % muscone, 0.81 wt % ethylene brassylate, 0.53 wt % phenyl hexanol, 0.51 wt % methyl salicylate, 0.35 wt % ethyl butyrate, and 0.33 wt % *cis*-3-hexenol, was prepared as an indicator of the perceived odor intensity. The indicator for weak was 0.80 wt % muscone, that for moderate was 0.81 wt % ethylene brassylate, that for strong was 0.53 wt % phenyl hexanol, those for very strong were 0.51 wt % methyl salicylate and 0.35 wt % ethyl butyrate, and that for the strongest imaginable was 0.33 wt % *cis*-3-hexenol. The panelists utilized these values while evaluating the odor intensity of PRM samples.

2.5. Modeling of Perceived Odor Intensity. In general, it is well-known that the shape of change in sensory strength relative to the logarithm of the stimulus in olfaction is quite close to an S shape, which fits well to a sigmoid curve, for example, Hill's law.^{1,30,31} The result of the obtained data showing the relationship between odor intensity and gas concentration of each PRM was plotted, and the curve was fitted with the logistic equation given by

$$\Psi_i = \frac{I_{\max,i}}{1 + e^{-(X_i - C_i)/D_i}} \quad (1)$$

where Ψ_i is the odor intensity and X_i is the logarithm of the concentration of the *i*th PRM in the gas phase ($\mu\text{g/L}$ air). Further, $I_{\max,i}$, C_i , and D_i are parameters specific to each PRM as follows: $I_{\max,i}$ is the odor intensity at saturated gas concentration, C_i is the value of X_i when Ψ_i is $I_{\max,i}/2$, and D_i is the steepness of the function. These parameters were determined using the solver function, which is an add-in program of Microsoft Excel 2010 (ver14.0) (see Table S2). It is possible to calculate the LMS value for each PRM from only arbitrary gas concentration by using eq 1.

Further, to calculate the odor intensity of mixtures consisting of various types of PRMs with different odor intensities and gas concentrations, the stronger component (SC) model^{32,33} was used. The theory of this model states that a mixture's odor intensity in the air tends to be close to the intensity of the PRM perceived and recognized as the strongest among the multiple components by the human nose. This model is expressed by the following equation:

$$\Psi_{\text{mix}} = \max\{\Psi_i\}, \quad i = 1, \dots, N \quad (2)$$

where Ψ_{mix} denotes the odor intensity of the mixture. Compared with other models for a mixture's odor intensity, such as the vectorial model, U model, UPL2 model, additivity model, equiratio mixture model, and Euclidean additivity model, the SC model is a simpler model because other parameters obtained from the experimental data are not required to predict the multicomponent odor intensity.^{32–35} The SC model has a simple principle in terms of predicting a mixture's odor intensity, and it guarantees that the intensity of a mixture will not be extremely large or small compared to the odor intensity of the strongest material among the multiple components. However, the model should be applied carefully in predicting the intensity of the mixture's odor because it cannot consider the synergistic or masking effect of odor interactions.

2.6. Verification of Calculation Model. To verify the accuracy of the prediction model developed in this study, the relationship between the odor intensity evaluated by the panels

and that predicted with the model was evaluated. A group of 11 subjects [8 males and 3 females; age range, 26–42 years; median age, 33.9 (male) and 33.0 (female)] comprising evaluators and perfumers actively engaged in fragrance evaluation and creation participated in the verification. All the subjects had the aptitude to pass the panel selection test with five standard odorants. In addition, to confirm the test-retest by the panelist, three PRMs out of the five standard odorants were dissolved in each liquid paraffin and evaluated before the verification of the model (Table S3 and Figure S1). One of subjects had been awarded a national qualification as an olfactory measurement operator by the Japan Association on Odor Environment designated by the Japanese Ministry of the Environment. Measurement of odor intensity was conducted by 3–5 panels randomly selected from the above-mentioned group in accordance with the Japanese simple olfactory test method to reduce the time and cost of measurement and to accumulate the data efficiently. For single-component verification, three PRMs, namely limonene, linalool, and methyl ionone-gamma, which differ in terms of scent lasting tendency, widely used to create fragrances and regarded as representative ingredients from the viewpoint of fragrance development, were selected. Then, their sample bags were prepared. For multicomponent verification, five types of fragrances composed of five perfume ingredients (see Table 2), which are regarded as important for constructing the

Table 2. Components of the Mixture

sample	compound	prescription ratio
(A)	limonene	36.0%
	linalyl acetate	41.5%
	linalool	12.0%
	beta-pinene	8.5%
	citral	2.0%
(B)	aldehyde c-12 mna	2.0%
	triplal	1.3%
	isobornyl acetate	51.3%
	nerolin yara yara	7.5%
	linalyl acetate	38.0%
(C)	aldehyde c-111 len	0.6%
	methyl benzoate	0.6%
	methyl ionone-gamma	8.5%
	coumarin	14.0%
	phenyl ethyl alcohol	76.4%
(D)	<i>cis</i> -3-hexenol	7.5%
	hexyl acetate	12.5%
	ethyl butyrate	3.0%
	o-t-b.c.h.acetate	75.8%
	alpha-damascone	1.3%
(E)	beta-ionone	52.0%
	gamma-decalactone	13.0%
	linalool	26.4%
	<i>cis</i> -3-hexenyl acetate	0.7%
	lilial	8.0%

character of the fragrances, were selected and the sample bags were prepared. These fragrances were created by perfumers working at Kao Corporation for this experiment with the following imaging character tone: sample (A) is bergamote, sample (B) is herbal, sample (C) is Chanel N°5, sample (D) is apple, and sample (E) is osmanthus.

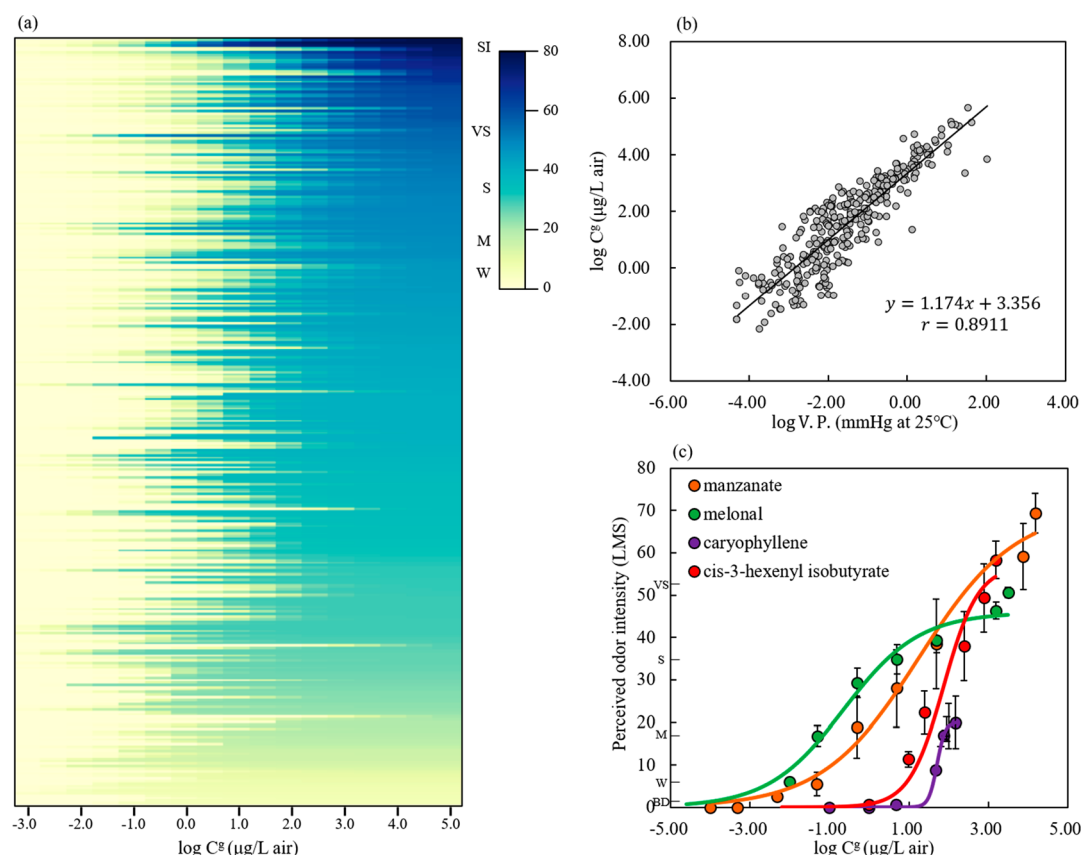


Figure 1. (a) Heat map of each PRM odor intensity. (b) Relationship between adequately dense PRM gas concentration inside sample bag measured with GC/MS and vapor pressure at 25 °C. (c) Odor intensity standard curve for each PRM.

RESULTS AND DISCUSSION

3.1. Odor Intensity Standard Curves (OISC). Here, information on the odor intensity corresponding to the concentration in the gas phase of each of the 314 PRMs was obtained.

Figure 1a shows a heat map providing information on the odor intensities of all the PRMs calculated using eq 1 within the gas concentration range, i.e., log C_g, from −5.0 to 5.0. By using the LMS as the evaluation scale of odor intensity, we can understand perceived intensity as not only a value (0–100) but also verbally (from barely detectable to strongest imaginable). Figure 1b shows the relationship between each PRM's vapor pressure estimated using software (ACD Percepta BP/VP V2016) and saturated gas concentration obtained from GC analysis. As the PRM vapor pressure is different for each material, the prepared gas concentration is also different. In summary, the limit of measurable maximum gas concentration in this measurement depends on the PRM's vapor pressure. For this reason, if the gas concentration is higher than the measured concentration used to calculate the LMS value from eq 1, the calculated value is programmed to output the same odor intensity as that evaluated at the highest gas concentration in the measurement (Figure 1a).

On the other hand, the odor intensity standard curve (OISC) shown in Figure 1c, which plots the relationship between odor intensity and gas concentration, can be described using the parameters obtained by curve fitting of the evaluation data with Microsoft Excel 2010. The important features of OISC are as follows: (1) providing information on PRM maximum odor intensity, slope, and threshold and (2)

estimating the LMS value from arbitrary gas concentration directly. Figure 1c shows that the property of odor intensity is different for each PRM. Here, PRMs with low threshold, such as manzanate and melonal, and others with high threshold, such as beta-caryophyllene and cis-3-hexenyl isobutyrate, are taken as examples. Even among PRMs with similar thresholds, the maximum intensity and slope are different. Therefore, melonal and cis-3-hexenyl isobutyrate show similar levels of maximum odor intensity, but each of them shows unique dose–response relationship: melonal continues to increase the odor intensity moderately by 6 orders of magnitude from threshold to maximum intensity, while cis-3-hexenyl isobutyrate attains a similar maximum intensity from the threshold with only 3 orders of magnitude. As a result, it is possible to describe the characteristic relationship between the odor intensity and the gas concentration of an individual ingredient by using OISC. Moreover, OISC can be used for threshold estimation. As the LMS value of 1.4 verbally denotes barely detectable, we considered that a human could perceive the odor if the value was greater than 1.4, which was regarded as the perception threshold of the gas concentration. The gas concentration (perception threshold) at LMS = 1.4 is calculated using the following equation:

$$Thr_i(\text{ng/L}_{\text{air}}) = 10^{(3+D_i - C_i \times \ln(\frac{I_{\text{max}_i} - 1.4}{1.4}))} \quad (3)$$

3.2. Consistency of OISC with Previous Observations.

For considering the reliability of OISC, 11 commonly used PRMs were selected from among the 314 PRMs for comparison with the detectable threshold reported in a previous study. The detectable threshold was used to verify

Table 3. Properties of PRMs

Cas no.	molecule (<i>i</i>)	physical property	odor property			threshold _{<i>i</i>}	
		vapor pressure, ^a (mmHg)	<i>I</i> _{maxi}	<i>C</i> _{<i>i</i>}	<i>D</i> _{<i>i</i>}	OISC (ppm)	reference ^b (ppm)
83–34–1	skatole	0.014	29.64	−2.26	0.35	0.0000924	0.0000056
503–74–2	isovaleric acid	0.518	54.37	−1.01	0.52	0.0002973	0.000078
106–44–5	<i>p</i> -cresol	0.196	51.68	−0.19	0.63	0.0008173	0.000054
105–54–7	ethyl butyrate	13.297	81.16	2.62	1.20	0.0008173	0.00004
120–72–9	indole	0.028	43.59	−0.19	0.54	0.0019906	0.0003
142–92–7	hexyl acetate	1.311	48.92	1.66	0.84	0.0081786	0.0018
111–87–5	<i>n</i> -octanol	0.106	40.11	0.60	0.51	0.0150324	0.0027
80–56–8	alpha-pinene	3.309	47.36	2.08	0.79	0.0389315	0.018
5989–27–5	limonene	1.455	52.62	1.71	0.50	0.1490639	0.038
18172–67–3	beta-pinene	2.271	44.33	2.08	0.60	0.1915940	0.033
140–88–5	ethyl acetate	107.589	35.06	2.95	0.35	18.85207945	0.87

^aCalculated using software (ACD Percepta BP/VP V2016). ^bFrom Nagata and co-workers.³⁶

the reliability for the following reasons. First, it is rather difficult to obtain data showing the PRM slope and maximum odor intensity because there are few reports on the relationship between odor intensity and gas concentration with LMS as the evaluation scale. Second, the comparison becomes more difficult even though the data are collected by various research groups conducting the measurement under different conditions, as the values of the slope and maximum odor intensity vary with the standard setting of the sensational scale. For comparison, it is preferable to use the data obtained previously under unified test conditions. Of course, regarding detection threshold, comparison is difficult because the value fluctuates considerably with the measurement conditions and it is not defined formally. Therefore, the threshold data obtained under the unified test conditions³⁶ were used in this case. Table 3 provides information on 11 materials used for comparison, vapor pressure, parameters of OISC, detection threshold estimated from OISC (concentration at LMS = 1.4), and that obtained in the previous study. Figure 2 shows the correlation

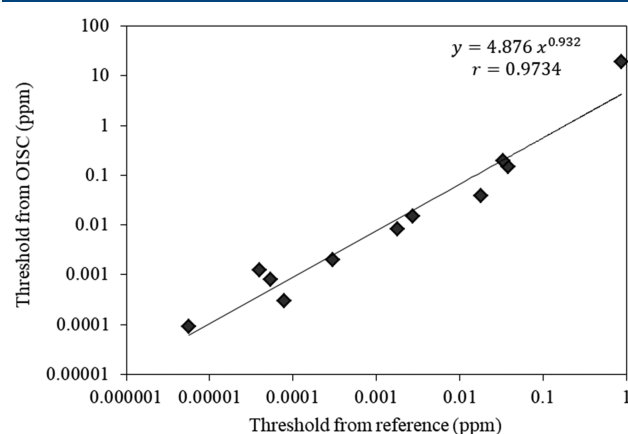


Figure 2. Correlation between threshold calculated from eq 3 and that obtained in the previous study.

with each threshold. The obtained correlation was comparatively high ($r = 0.98$). Although the threshold calculated from eq 3 cannot be officially established, this result shows that these data are in good agreement with the results of the previous study and the OISC data obtained in this study are relatively valid.

3.3. Comparison between Evaluations and Predictions for a Single Component.

Here, we investigated

whether the OISC model can predict the odor intensity of the newly recruited subjects. In this experiment, three representative PRMs, namely, limonene, linalool, and methyl ionone-gamma, were selected; they are widely used for fragrance creation from the viewpoint of fragrance development and have different tendencies of lasting scent in the top note, middle note, and last note. Then, three or four sample bags, each with a different gas concentration of a single component, were prepared. Subsequently, these sample bags were connected to FAS-1, and the sample gas diluted to various concentrations inside FDL-1 was discharged from the Teflon tube. The root-mean-square error (RMSE) was confirmed from the experiment in which the subjects evaluated the odor intensity of 16 concentrations in limonene, 16 concentrations in linalool, and 14 concentrations in methyl ionone-gamma. RMSE is one of the statistical parameters showing the magnitude of prediction error.³⁷ A value of RMSE close to 0 means that the model prediction accuracy is quite high. Figure 3a–c shows the results of evaluating and predicting odor intensity for corresponding gas concentrations of each of the three PRMs. The RMSEs of the evaluated and predicted values for limonene, linalool, and methyl ionone-gamma were 4.56, 7.99, and 6.11, respectively. The average of the RMSEs of the three PRMs was 6.22 (Figure 3d and Figure S1). The RMSE values for the three PRMs and detailed comparisons between prediction and evaluation are given in Table 4 and Table S3. To confirm the precision of the accuracy of the model, we also calculated the RMSE values of predictions based on the OISCs of mismatched PRMs (e.g., the evaluated intensity of limonene was predicted by OISC of linalool). The average of 18 RMSE values from mismatched OISC-based predictions was 12.85, which is higher than the 6.22 obtained from the correct models (Table 4 and Figure S1). This result demonstrated the accuracy of OISC-based predictions of the odor intensity of the newly recruited panelists. However, the result also highlighted a degree of error in prediction when comparing the predicted value and measured values for each PRM (Figure 3a–c). This error may be attributed to the difference in the members of the panel who evaluated the respective concentration of PRMs and who differed from each other in genetic background and internal states.^{38,39} Indeed, our panelists showed a significant level of across-subject variability as well as within-subject variability in the control experiments (Figure S2 and Table S4). In conclusion, although a degree of error was observed, the odor intensity predicted by eq 1 from

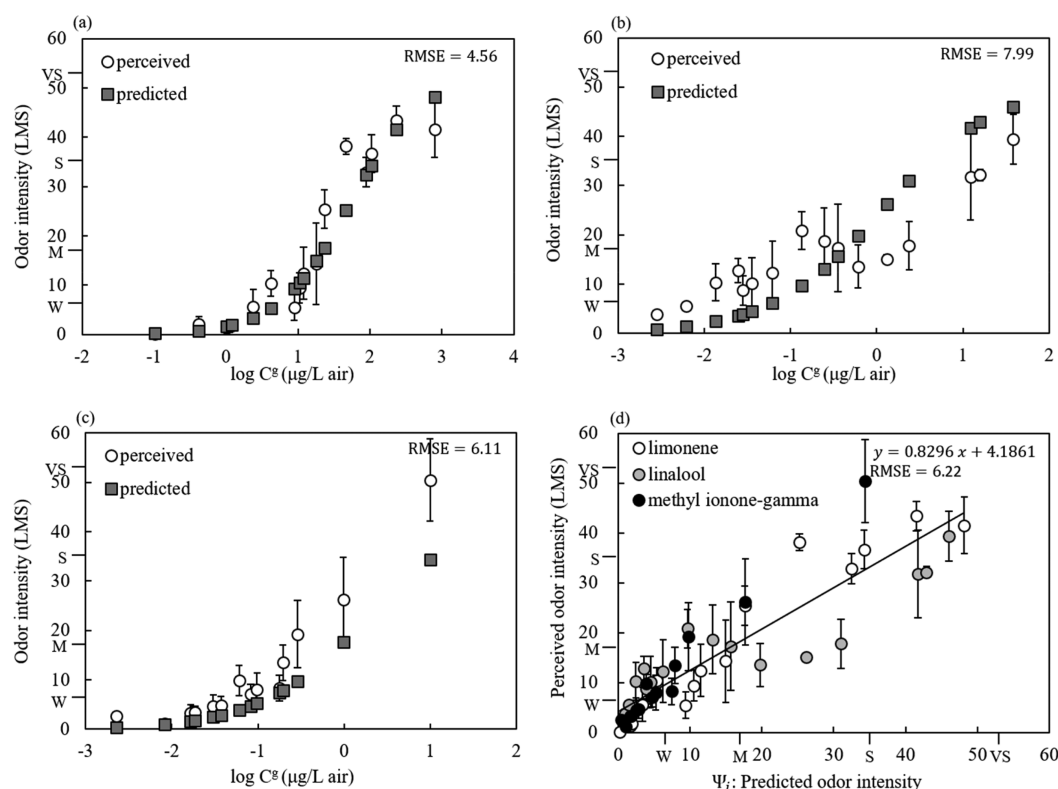


Figure 3. Comparison between perceived odor intensity and predicted odor intensity (Ψ_i) of (a) limonene; (b) linalool; (c) methyl ionone-gamma; and (d) the average of root-mean-square errors in this experiment; standard error, $n = 3-4$.

Table 4. Root Mean Square Errors of the Evaluated and Predicted Odor Intensity

		odor intensity standard curve parameters used in predictions						
		limonene	linalool	methyl ionone-gamma	manzanate	caryophyllene	cis-3-hexenyl isobutyrate	Melonal
component evaluated	limonene	4.56	21.52	16.18	15.81	15.07	6.46	22.26
	linalool	14.17	7.99	7.27	4.55	18.87	15.48	7.97
	methyl ionone-gamma	14.86	3.36	6.11	6.04	17.20	15.75	8.54

the gas concentration is comparatively close to the perceived odor intensity in practice.

3.4. Comparison between Evaluations and Predictions in Mixture. The main purpose of this experiment is to verify how accurately the model developed with OISC can predict the perceived odor intensity in the case of multiple components. First, some sample bags containing five types of fragrances listed in Table 2 (mixtures A–E), which are commonly used to construct basic scent tone for product fragrances, were prepared with adequately high gas concentration. These bags were connected to FDL-1 and diluted 1000-fold at 10-fold intervals. From the result of comparing 20 samples on the basis of the evaluation scores reported by the subjects and the prediction score calculated in accordance with Eq. 1 and SC model (Eq. 2), the statistical correlation between evaluation and prediction was confirmed. The results are shown in Table S5 and Figure 4. To confirm the accuracy of the prediction based on the OISC of the strongest component, the RMSE of the odor intensity of each mixture was calculated [mixture (A): 7.86, (B): 7.05, (C): 7.12, (D): 3.80, (E): 7.61] and their average was 6.69. This is smaller than most of the RMSE values calculated from the mismatched models in which the odor property of five components of each mixture was exchanged randomly with those of other components (an

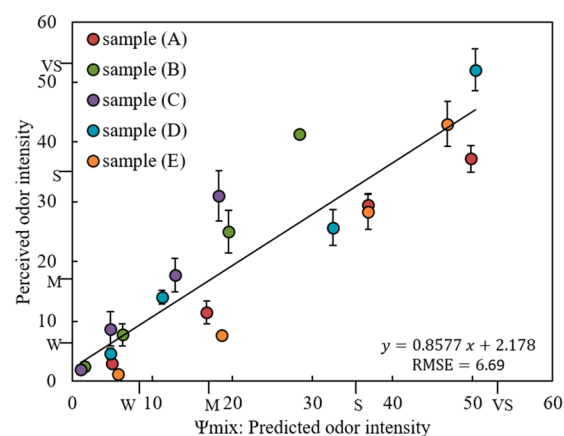


Figure 4. Correlation between perceived and predicted odor intensities of a mixture; standard error, $n = 3-5$.

average of 9.55 for five mixtures $\times$ five cases; Table S6 and Table S7). However, this series of experiments also highlighted the drawback with the application of the SC model. Previous studies have reported cases where a component showing smaller odor intensity value contributed to the perceived intensity, making it difficult to predict odor intensity based

only on the strongest component.^{40,41} Consistently, our analysis indicated that there were cases where prediction based on the strongest component of a mixture was less accurate than that based on PRMs which were not components of the mixture (Table S7). For example, the odor intensity of the mixture (B) was predicted with RMSE of 7.05 based on the OISC of the strongest components, whereas it was predicted more accurately with RMSE of 1.10 by the OISC of PRMs which were not components of the mixture (B).

In conclusion, this result shows that the SC model with eq 1 cannot always precisely predict the mixture's odor intensity because of the influence of the synergistic and masking effects in multicomponents⁴² (Figure S2). The application of OISC database, however, is expected to be one of the useful concepts for predicting the odor intensity of mixtures from arbitrary gas concentrations during fragrance development because there are some cases showing good prediction results (Figure 4).

CONCLUSION

A database of 314 PRMs widely used for fragrance development was developed to (i) establish a relationship between odor intensity and gas concentration and (ii) develop an odor intensity prediction model and thus establish a scientific foundation for activities requiring artistic and sensational skills, e.g., fragrance creation. The contributions of this study are as follows: (1) data covering a large number of commonly used PRMs were accumulated and were shown to agree with the results of previous research; (2) values quantifying 314 PRM odor intensities were calculated only from arbitrary gas concentrations; and (3) results showing that odor intensity can be directly predicted from the calculated values for both single and multiple components. This database allows the prediction of the odor intensities of the 314 PRMs used in this study, although the model should be applied carefully when predicting the odor intensity of a mixture because this model cannot consider the synergistic or masking effects of odor interaction.

The results of this study are expected to be of significance to the F&F industry. For example, by combining the developed model with the results of PRM behavior analysis and data on PRM vaporization or adsorption on skin, hair, and towels/textiles, one can finally realize a technology for predicting odor intensity from PRM prescription to products. If the fragrance character is scientifically elucidated and objectively explained, it would be possible to efficiently pass on long-term experience-enhanced skills to the next generation of perfumers and thus expedite their training. Moreover, the identification of PRMs not significantly contributing to the scent performance of a given prescription should allow one to decrease the number and quantity of chemical substances in this prescription, which is expected to facilitate the realization of a more sustainable society. Additionally, the results of this work are expected to contribute to the analysis of the mechanism of olfaction. Although the mechanism of human odor perception remains unclear, analysis of the relationship between odor sensation-related parameters and the response of olfactory receptors would accelerate the elucidation of the basic mechanism of odor sensing. The revelation of the synergistic or offset effect of odors based on the knowledge of olfactory receptors can be linked to a new approach for the development of fragrances with large-scale diffusion or effective masking of malodors through combinations of some PRMs.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.9b01225.

The RSD of each PRM with GC/MS (Table S1), odor properties of the PRMs (Table S2), RMSE of each single component for both the unshuffled and shuffled parameter (Table S3 and Figure S1), ability of each panelist to make stable evaluations (Table S4), probability of differences given the different genetic makeup or experience of the panelists (Table S4 and Figure S2), predicted odor intensity of mixture sample and average grade of perceived odor intensity (Table S5), shuffled parameter with other PRMs under five different conditions (Table S6), the RMSE of both unshuffled and shuffled odor properties in the SC model (Table S7), and the case that does not fit the SC model (Figure S3) (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. H.W. designed the research, conducted experiments, and analyzed the data. H.W., M.S., and K.Y. wrote the paper. M.I. supervised the research. All authors reviewed and edited the manuscript. All authors have given approval to the final version of the manuscript.

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Notes

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